

The Program in Action

When you run *ninetyninebottles.py*, the output will look like this:

Ninety-Nine Bottles, by Al Sweigart al@inventwithpython.com

```
(Press Ctrl-C to quit.)
99 bottles of milk on the wall,
99 bottles of milk,
Take one down, pass it around,
98 bottles of milk on the wall!

98 bottles of milk on the wall,
98 bottles of milk,
Take one down, pass it around,
97 bottles of milk on the wall!
--snip--
```

How It Works

The repetition in this song makes it easy to use a for loop (from lines 20 to 30) to display the first 98 stanzas. However, the last stanza has some minor differences and requires separate code to display (lines 33 to 39). This is because the last line, 'No more bottles of milk on the wall!', deviates from the line repeated in the loop, and because the word “bottle” is singular rather than plural.

```
1. """Ninety-Nine Bottles of Milk on the Wall
2. By Al Sweigart al@inventwithpython.com
3. Print the full lyrics to one of the longest songs ever! Press
4. Ctrl-C to stop.
5. View this code at https://nostarch.com/big-book-small-python-projects
6. Tags: tiny, beginner, scrolling"""
7.
8. import sys, time
9.
10. print('Ninety-Nine Bottles, by Al Sweigart al@inventwithpython.com')
11. print()
12. print('(Press Ctrl-C to quit.)')
13.
14. time.sleep(2)
15.
16. bottles = 99 # This is the starting number of bottles.
17. PAUSE = 2 # (!) Try changing this to 0 to see the full song at once.
18.
19. try:
20.     while bottles > 1: # Keep looping and display the lyrics.
21.         print(bottles, 'bottles of milk on the wall,')
22.         time.sleep(PAUSE) # Pause for PAUSE number of seconds.
23.         print(bottles, 'bottles of milk,')
24.         time.sleep(PAUSE)
25.         print('Take one down, pass it around,')
26.         time.sleep(PAUSE)
```